Intermediate Band Solar Cells: Significance of Intermediate Band Mobility

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(Dated: December 5, 2025)

Intermediate band solar cells (IBSCs) present an advantageous distinction from the conventional p-n junction solar cell, surpassing the Shockley-Queisser (SQ) efficiency limit [1]. The carrier mobility in the intermediate band (IB) has the potential to be a significant constraint in achieving optimal efficiency. Since there is no current produced directly by the IB, the necessity of carrier mobility in the IB is unclear. To achieve optimal efficiency, the carriers excited into the IB from the VB must be equal to the carriers excited from the IB to the CB. The excitation rate can vary greatly in different locations, providing the need for a transport method to equalize excitation rates. This document highlights the situations in which the intermediate band mobility is influential to the maximum efficiency of the solar cell.

I. INTRODUCTION

A. Efficiency Limit

The SQ limit prescribes a maximum efficiency of 31% for single p-n junction solar cells under one-sun unconcentrated blackbody radiation from an object at 6000K (approximate temperature of the sun) and 40.7% under fully concentrated blackbody radiation [1]. These limitations have since been refined, but with similar values. This limit is a common benchmark for technologies aimed to rival conventional solar cell efficiency. Seeking to surpass this efficiency, IBSCs implement a third energy band between the valence and conduction band of the solar cell material. This allows for the absorption of lower-energy photons without compromising the open-circuit voltage. A detailed balance analysis of IBSCs, conducted by Luque and Marti (LM), limits the achievable efficiency of an IBSC to 46.8% under unconcentrated blackbody radiation and 63.2% under fully concentrated blackbody radiation [2]. IBSCs present an efficiency which significantly surpasses the SQ limit, potentially revolutionizing the solar cell industry.

Detailed balance limits make significant approximations and idealizations to achieve a general efficiency limit. The LM limit assumes perfect absorption, purely radiative recombination, and infinite carrier mobility within the material [2]. In physical scenarios, there are several recombination mechanisms at play, the absorption of each region varies significantly, and the mobility of carriers can be far less than ideal. The effects these parameters have on the efficiency of IBSCs are not well understood and have the potential to hinder the practical implementation of IBSCs if neglected when engineering the IBSC materials [9].

B. Intermediate Band Mobility

As no carriers are directly extracted from the intermediate band, electrons excited to the intermediate band must then be excited to the conduction band in order to contribute to the power generation of the solar cell. Within the intermediate band electrons have the potential to drift or diffuse to other areas of the intermediate band but are limited by the carrier mobility of the band.

In some cases, the generation rate from the VB to the IB is larger than the generation rate from IB to CB, or vice versa. This discrepancy in generation rate is known as a mismatch and can occur both locally within certain areas of the band and globally across the IB. In the case of local mismatch, the electrons must be able to move through the material to other locations to balance local transitions. The research conducted within this report will discern when the mobility of carriers within the intermediate band has a substantive effect on the achievable efficiency of an IBSC.

C. Simudo

The IBSC band structure seen in Figure 3 presents a complex dynamic that can be difficult to resolve. The depletion regions at the boundaries within the cell are governed by the drift-diffusion equation and Poisson's equation, leading to a difficult differential equation that must be solved numerically [7].

To model this complex system, the python package Simudo facilitates the simulation of an IBSC using a mesh framework and solving the coupled Poisson and drift-diffusion equations throughout the device [7]. This model also incorporates Shockley-Read-Hall (SRH) recombination, surface recombination, trapping and the potential for additional recombination models such as Auger re-

combination. Through the Simudo software, the absorption coefficients, fill fraction and carrier mobility in each region are also spatially variable parameters. These can each be modified to observe their impacts.

II. THEORY

A. PN Junction

Before delving in to the complexities of IBSCs, an understanding of the limitations of conventional solar cells can be very useful. Beginning with a generalized semiconductor material consisting of an atom with a valency of four, each atom forms a covalent bond with four other atoms in a perfect crystal. Note that there are numerous combinations of atoms with different valencies that can be used, but valency four atoms are the more common.

The materials crystal structure has two main energy bands, the valence band (VB) and conduction band (CB) separated by a bandgap E_g. Each band has a number of allowed discrete states that electrons can occupy the number of these states per unit volume is known as the density of states. The number of electrons within each band at a given temperature is given by the Fermi-Dirac distribution [3]:

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1}. \tag{1}$$

Here, E is the energy of an allowed electron state, k is Boltzmann's constant, T is the temperature of the material and E_F is the Fermi energy. The Fermi energy defined as the energy at which the probability of occupancy is 0.5. For an intrinsic semiconductor with no impurities at room temperature, the Fermi energy is located halfway between the VB and CB.

If an atom with a valency of five is introduced into this crystal, then it will produce an additional electron that will not form a bond and will act as a nearly-free electron within the CB, loosely bound by the nucleus of the impurity. Similarly, if an atom of valency three is introduced, then it will leave an additional vacancy in the VB near the impurity site [3]. In semiconductor physics, the empty space left by an excited electron within the VB is known as a hole. Although holes are not physical particles, they can be treated as such, with an effective mass and charge assigned to them. The introduction of holes greatly simplifies the conceptualization of semiconductor physics.

Semiconductors are doped with these two types of impurities to form what are known as n-type and p-type materials, so named because of the additional negative and positive carriers respectively [3]. An example of impurities, also known as dopants, being introduced into a silicon lattice is seen in Figure 1. Boron provides the p-type dopant and Phosphorous provides an additional electron to introduce n-type doping.

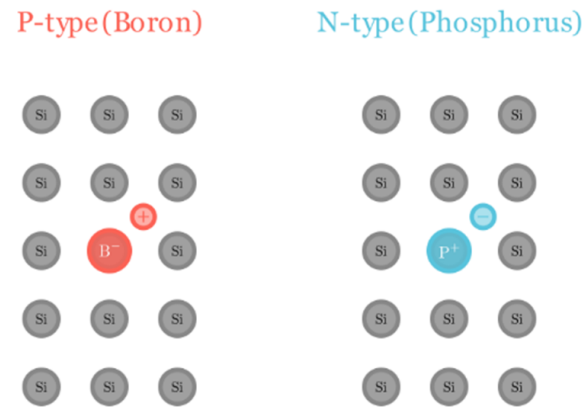

FIG. 1. P-type (Boron) and N-type (Phosphorus) doping in Silicon.

Within these n-type and p-type materials, the Fermi energy of the material is altered locally by the addition of holes or electrons [3]. This shifting is caused by an increase in the number of holes in the VB or electrons in the CB, thus shifting the Fermi-Dirac distribution to center around a new position. This creates a global change in the effective Fermi energy, shifting the effective Fermi energy higher or lower depending on the type of doping. Additional electrons shift the Fermi energy of the material higher while additional holes shift the energy lower.

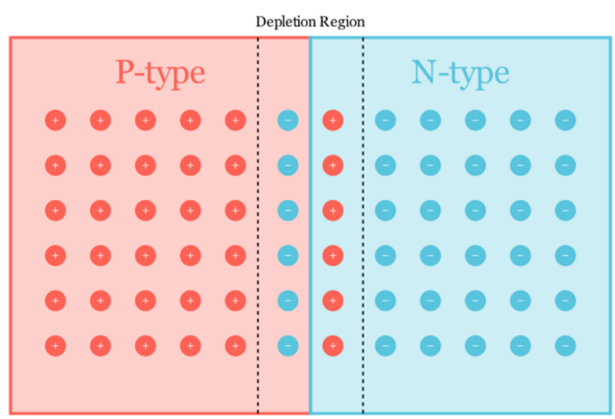

FIG. 2. Formation of the Depletion Region at a PN Junction.

When these two materials are abruptly introduced at some boundary as seen in Figure 2, they form what is known as a p-n junction [3]. The boundary of the device allows the overabundance of carriers in each material to diffuse into the other and recombine. As the carriers

diffuse, they leave behind ionized atoms. The exchange of carriers creates what is known as a depletion region around the boundary, as can be seen in Figure 2. The ionized atoms within the depletion region introduce a variation in the charge density, producing an electric field as prescribed by the Poisson equation:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}. \tag{2}$$

This electric field opposes the diffusion of carriers and causes them to drift against the diffusion current. On the molecular scale this is a fairly chaotic process, but macroscopically the carriers appear to move at a single velocity known as the drift velocity $\vec{v}_d$ [3]:

$$|\vec{v}_d| = |\mu \vec{E}|. \tag{3}$$

Here μ is the carrier mobility and is dependent on the impurity concentration of the material. The resulting current density of carriers created by this drift is described by the drift-diffusion equations [3]:

$$\vec{J}_n = qn\mu_n \vec{E} + qD\frac{dn}{dx}, \quad \vec{J}_p = qp\mu_p \vec{E} - qD\frac{dp}{dx}. \tag{4}$$

Here $\vec{J}$ is the current density, q is the fundamental charge, n and p are the carrier concentrations, μ is the carrier mobility for each type of carrier, and D is the diffusion coefficient [3]. These equations eventually reach an equilibrium where $\vec{J} = 0$ although the drift and diffusion terms are non-zero. The equations are essential to the operation of a solar cell and provides the foundation for the current produced by a solar cell.

B. Conventional Solar Cell

The cornerstone of solar cell physics is the photovoltaic effect, a close derivative of Einsteins Nobel-winning photoelectric effect. The photovoltaic effect dictates that an incident photon, with an energy greater than the band gap of a semiconductor material, can interact with and excite an electron from the valence band (VB) to the conduction band (CB) [4]. This will produce what is known as an electron-hole pair as there will be a hole left in the VB as the electron is excited.

When a photon flux is introduced to the p-n junction, the excited electrons are forced towards the n-type side by the electric field, and the holes are forced towards the p-type material [4]. This increase in carriers increases the diffusion current, counteracted by a growing electric field, maintaining the drift-diffusion equation at a steady-state with $\vec{J} = 0$.

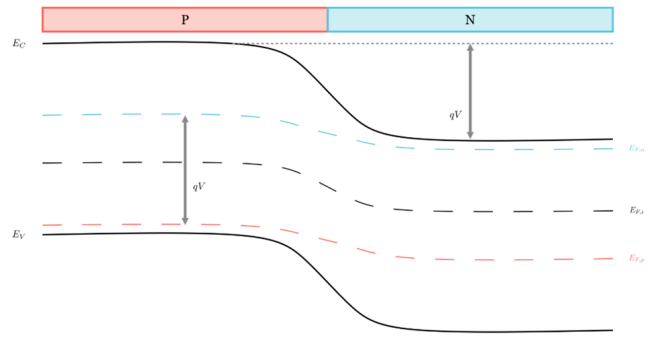

FIG. 3. Band diagram demonstrating the splitting of quasi-Fermi levels ($E_{F,n}$ and $E_{F,p}$).

The increased number of electron-hole pairs creates a difference in the effective Fermi-level seen by the electrons and holes seen in Figure 3. Conceptually, there will be an increase in holes in the VB, meaning that the Fermi-Dirac distribution must accommodate the increased probability of carriers in the VB. Likewise, electrons in the CB become more populated, leading to a splitting of the Fermi level into two quasi-Fermi levels as can be seen in Figure 3 with $E_{F,n}$ and $E_{F,p}$ [4].

The splitting of the quasi-Fermi levels creates a voltage potential across the depletion region which can be measured equivalently by two metrics seen in Figure 3 [4]. The difference in the conduction band potential across the depletion region, or the difference in the quasi-Fermi levels given by:

$$qV = E_{F,n} - E_{F,p}. \tag{5}$$

By attaching metallic contacts to each end of the p-n junction, this voltage potential can be harnessed to produce a current. The electrons are allowed to flow through some conductive circuit and recombine with the holes in the p-region. As the resistance of the load is reduced, the current increases, reducing the number of available carriers [4]. This process also reduces the potential across the junction as the quasi-Fermi levels grow closer. The optimal power created by a solar cell is a curated balance of the junction potential and the net current density of the cell.

The number of electron-hole pairs generated within the solar cell is dependent on several factors, the most obvious of which is the photon flux into the solar cell. The generation rate is also dependent on the energy of the incident photons, as only photons with an energy higher than the band gap will be absorbed [4]. We consider an incident solar flux modeled by the blackbody spectrum of an object at the temperature of the sun:

$$E(\lambda, T) = \frac{2hc^2}{\lambda^5(e^{hc/\lambda kT} - 1)}. \tag{6}$$

In a perfectly absorptive solar cell, every photon with energy greater than the band gap of the semiconductor

is absorbed [4]. The generation rate is then described by

$$\dot{N}^{absorbed} = \frac{\pi}{h^3c^2}\int_{E_g}^{\infty}\frac{E^2}{e^{E/kT_{sun}}-1}dE. \quad (7)$$

This is a crucial limiting factor in solar cell efficiency [2]. There are also several systems by which electrons in the conduction band recombine with holes in the valence band, limiting the total number of excited electrons [3]. The most prevalent case of recombination is radiative recombination, whereby a photon is emitted by an electron in the CB, and the electron recombines with a hole in the VB. The rate of thermal recombination is described by [2]:

$$\dot{N}^{emitted} = \frac{\pi}{h^3c^2}\int_{E_g}^{\infty}\frac{E^2}{e^{(E-qV)/kT_{cell}}-1}dE. \quad (8)$$

When considering only radiative recombination, the current density through the solar cell is given by:

$$J = q(\dot{N}_{CV}^{abs} - \dot{N}_{CV}^{emit}), \quad (9)$$

where q is the elementary charge.

C. Detailed Balance

A detailed balance limit is an analysis of the maximum efficiency a solar cell can produce. We consider a conventional solar cell with perfect absorption, only radiative recombination, and infinite carrier mobility. In this model, the voltage and current output of the solar cell each vary with the band gap energy, which can be manipulated to achieve the maximum efficiency for some incident spectrum of light.

The detailed balance limit of conventional, single junction solar cells, first described by the SQ limit, limits the maximum efficiency of a conventional solar cell to 31% under one-sun radiation and 40.7% under fully concentrated blackbody solar illumination [2].

Beyond the detailed balance limit, there are many other factors that limit the efficiency of the solar cell. There are several other forms of recombination that can significantly reduce the efficiency of a solar cell. Trapping is a common issue, when trapping sites such as impurities are introduced that have local state within the band gap, they "trap" electrons locally [3]. Surface recombination is caused by large numbers of impurities present at the surface of the material and capture electrons similarly to the trapping mechanism. There are several other forms of recombination, each of which contribute to the total recombination rate and reduce the efficiency of the material.

An additional limitation of a solar cell is the mobility of the material, reduced mobility means that in order to achieve the same drift velocity, a larger electric field is required as prescribed by equation (3). The mobility of the material is dictated by the rate of recombination, along with the filling fraction of the band. If a band is filled with electrons, then the rate of collisions increases, and the drift velocity also decreases.

D. Intermediate Band Solar Cells

To surpass the SQ limit, the structure of the solar cell can be altered to improve the absorption of the solar spectrum. The introduction of an intermediate band (IB) between the conduction band (CB) and valence band (VB) allows the solar cell to absorb lower energy photons without compromising the OC voltage [2]. The band structure prescribed by an IBSC can be seen in Figure 4, wherein three band energies, E_C, E_I and E_V can be seen.

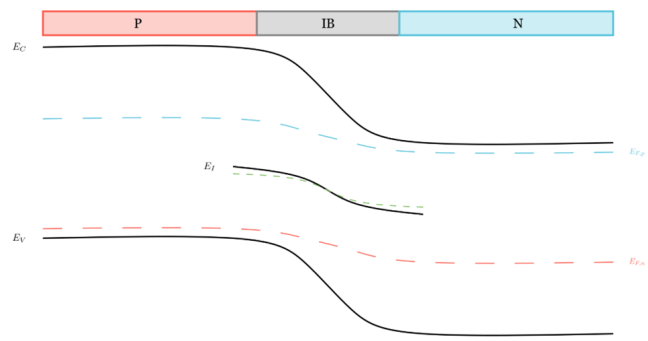

FIG. 4. Intermediate Band Solar Cell band structure showing E_C, E_I, and E_V.

The conventional solar cell can only absorb photons with an energy greater than its band gap energy, limiting their efficiency significantly [2]. To circumvent this, IBSCs incorporate an additional band between the CB and VB. This allows electrons in the valence band to be excited to the conduction band by two consecutive photons, one photon with energy greater than $E_I - E_V$ and a second photon with energy greater than $E_C - (E_I + E_V)$. This allows the solar cell to collect a larger portion of the solar spectrum without compromising the OC voltage.

The current density provided by an IBSC, considering only radiative recombination and perfect absorption, is calculated with the available photon fluxes [2]:

$$J = q(\dot{N}_{CV}^{absorbed} + \dot{N}_{CI}^{absorbed} - \dot{N}_{CV}^{emitted} - \dot{N}_{CI}^{emitted}) \quad (10)$$

Similarly to the SQ limit, the LM limit is a detailed balance limit of IBSCs, producing a maximum efficiency of 46.8% under unconcentrated sunlight and 63.2% under fully concentrated sunlight [2]. This limit presents a sub-

stantive potential for IBSCs to surpass the efficiency of conventional solar cells in practical application.

Creating a detailed balance limit of an IBSC involves balancing both the generation rate and the radiative recombination rate to achieve optimal efficiency. This is done by modifying the CB energy, the IB energy and the quasi-Fermi levels to obtain the optimal efficiency. The heatmap presented in Figure 7, similar to the ones on [6], demonstrates the relationship between the maximum efficiency of different combinations of E_I and E_g.

III. PROBLEM: INTERMEDIATE BAND MOBILITY

The LM limit assumed infinite mobility in all bands [2], however, in practical IBSCs, the intermediate band has an increased non-radiative recombination rate because of the additional impurities. Because of this increased recombination rate, the mobility of the intermediate band can be significantly impacted. This reduced mobility could pose significant barriers in achieving the optimal efficiency of a practical solar cell.

Consider the case of an IB located below $\frac{1}{2}E_g$, where E_g is the band gap depicted in Figure 5 [9]. If we only allow photons with energy E_{IV} to E_{CI} to be absorbed by the first transition and photons with energy between E_{CI} E_g to be absorbed by the second transition, and we assume perfect absorption then the flux of electrons generated will be:

$$\dot{N}_{IV} = \int_{E_{IV}}^{E_{CI}} \Phi_0(E)dE, \quad \dot{N}_{CI} = \int_{E_{CI}}^{E_g} \Phi_0(E)dE \quad (11)$$

Note that the bounds of integration and subscripts must be flipped for the case where $E_I > \frac{1}{2}E_g$ ($E_{CI} < E_{IV}$):

$$\dot{N}_{CI} = \int_{E_{CI}}^{E_{IV}} \Phi_0(E)dE, \quad \dot{N}_{IV} = \int_{E_{IV}}^{E_g} \Phi_0(E)dE \quad (12)$$

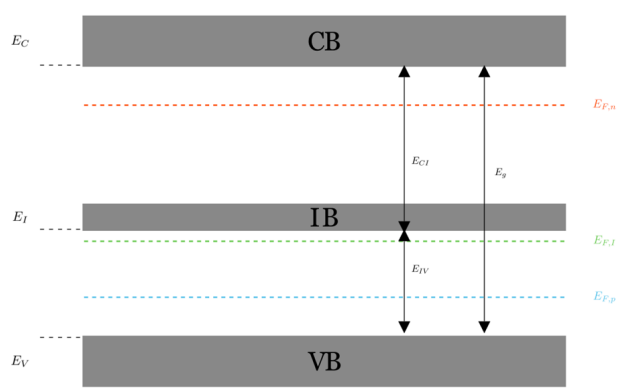

FIG. 5. Energy transitions and quasi-Fermi levels in an IBSC.

Next, we introduce a parameter to quantify the difference in generation rate between the two transitions. The flux values can then be used to characterize a value deemed the global current mismatch in [9]:

$$m_G = \frac{|\dot{N}_{CI} - \dot{N}_{IV}|}{\dot{N}_{CI} + \dot{N}_{IV}}. \quad (13)$$

This global current mismatch defines how disproportionate the availability of photons is between the two transitions is. When the global current mismatch is zero, the solar cell is able to achieve the maximum efficiency at the defined CB and IB energy [9].

Practically speaking, there is no such thing as perfect absorption. Instead, we prescribe each intermediate band absorption process with an absorption coefficient, α_{IV} and α_{CI} as in [9]. Each of these coefficients will determine the availability of each process to absorb photons. These coefficients will be zero below their respective band gap energy, and will otherwise follow:

$$\alpha_{IV}(E) = N_I \sigma_{IV}^{opt}(E)(1 - f_I), \quad (14)$$

$$\alpha_{CI}(E) = N_I \sigma_{CI}^{opt}(E) f_I. \quad (15)$$

Here f_I is the filling fraction, a ratio of the electron density to the density of free states within the intermediate band and σ^{opt} is the optical capture cross-section [9].

Current mismatch is not solely a global factor, the absorption coefficients are a function the incident photon flux, which decreases deeper within the cell. Therefore, at a position x within the intermediate band, the local current mismatch can be defined as in [9]:

$$m_L = \frac{\int (g_{IV}(x) - g_{CI}(x))^2 dx}{\int (g_{IV}(x) + g_{CI}(x))^2 dx}, \quad (16)$$

Where $g(x)$ is the net flux of electrons for each process. At each location x within the intermediate band, the generation rate is defined by [9]:

$$g_{IV}(x) = \int_{E_{IV}}^{E_{CI}} \Phi(x, E)\alpha(x, E)dE. \quad (17)$$

Here is where the mobility of the intermediate band becomes crucial. If the global current mismatch is minimal, then the two processes should absorb an equal amount of photons, however if there is a local mismatch within the IB, there must be a mechanism by which the electrons can be transported to the location of the local mismatch from a different location to balance the transitions [9]. If the mobility is to little within the IB, then the electrons will recombine before they are able to move towards or away from the local mismatch. This inability to satisfy one of the transitions at a certain location within the IB will reduce the efficiency of the solar cell by limiting the total carrier flux supplied by the IB.

IV. PREVIOUS RESEARCH

Research into the effects of intermediate band mobility have produced mixed results. First, in Simudo: a device model for intermediate band solar cells [7], the Simudo software is used to investigate the effects of low intermediate band mobility. The results demonstrate that when the optical absorption is well matched in both the valence to intermediate (IV) process and the intermediate to conduction (CI) process, the efficiency of the device is relatively unaffected by any variation in IB mobility. However, when the optical absorptions of the two processes are not well balanced, the intermediate band mobility can have a significant impact on the efficiency of the device. Notably this simulation was performed with a constant mobility model, meaning the carrier mobility within the IB remains constant.

To expand on this observation, the research performed in Material Quality Requirements for Intermediate Band Solar Cells [8] better demonstrates the effects of the IB mobility. Not only does it provide more in-depth results, but it also implements a different mobility model (although not mentioned in the text) which incorporates the filling fraction into the mobility. The carrier mobility in the intermediate band is modeled by:

$$\mu = \mu_0[(1 - f_I) - \frac{1}{2}(f_I^{\frac{1}{3}}(1 - f_I)^{\frac{4}{3}})]. \tag{18}$$

Here the parameter μ_0 is the baseline mobility specified for the material and is used to modify the amplitude of the mobility. The equation provides a general expected form of the mobility and will decrease as the intermediate band fills.

The results of this study further corroborate the significance of intermediate band mobility in the efficiency of IBSCs. By analysing the efficiency of the IBSC over three different variables, a fairly well-defined pattern was produced. The simulation was set up with an intermediate band gap below $E_g/2$ and found that when the optical cross-section of the IV process was significantly greater than that of the CI process, the impacts of IB mobility were significant. In other cases, with equal or lower σ_{IV}^{opt} the IB mobility had a negligible impact on the efficiency of the solar cell. Each of these simulations used a band gap energy of $E_g = 1.67\ eV$, in order to thoroughly demonstrate that this IB mobility dependence holds for all IBSCs, the effects must be test at multiple band gap energies.

V. RESULTS

A good deal of the fall semester involved gaining a thorough understanding of the complexities of intermediate band solar cells. Beginning with an exploration of the basic limitations of conventional solar cells and IBSCs. First, in September I created a program to model the SQ limit. This was fairly straightforward and produced the plot in Figure 6. The plot demonstrates the 31% efficiency limit of conventional solar cells, along with a fairly sharp peak in the optimal efficiency.

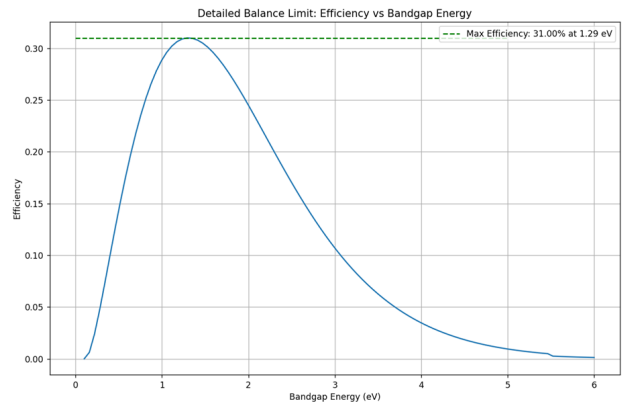

FIG. 6. Detailed Balance Limit: Efficiency vs Bandgap Energy.

The optimizer iterated over a list of band gap energies and used each value to optimize the applied voltage across the cell from equation (5). This in turn was optimizing the difference in quasi-Fermi levels within the solar cell.

Next, during October, a more complex detailed balance limit was in order to replicate the results of LM. This simulation proved difficult as there was more than one variable to optimize. In order to simulate the IBSC limit, a similar optimizer was constructed that took two lists of E_g and E_I values plotted in Figure 7, and determined the optimal quasi-Fermi level for all three quasi-Fermi energies. The plot in Figure 7 also depicts the maximum efficiency of the solar cell with the black dot at $\sim E_g = 1.97$ eV and $\sim E_I = 0.72$ eV.

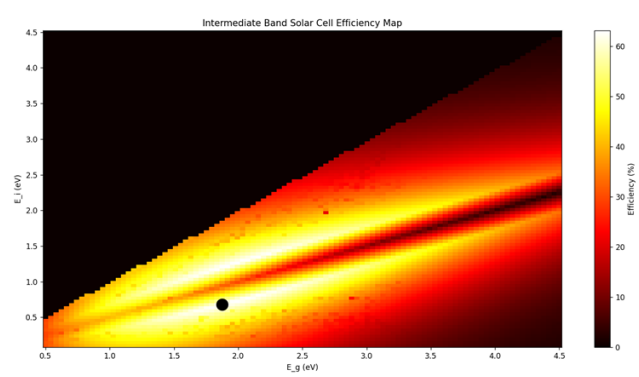

FIG. 7. Intermediate Band Solar Cell Efficiency Map.

The simulation demonstrated the bimodal efficiency peaks of the optimizer, with the most efficient E_I values presented on either side of $E_g/2$. Additionally, the peak efficiency using solely a list of E_g values was produced in Figure 8 using a two-dimensional optimizer. This figure also includes the peak efficiencies at one-sun solar concentration.

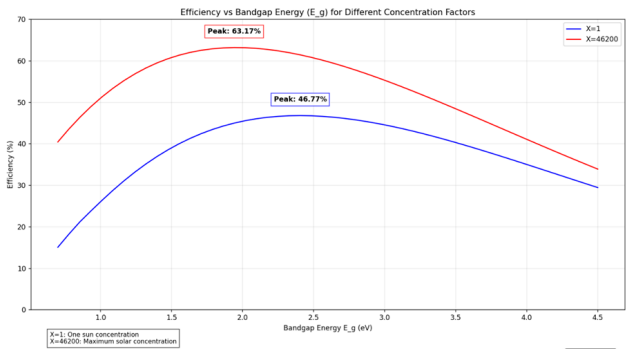

FIG. 8. Efficiency vs Bandgap Energy (E_g) for Different Concentration Factors.

After completing the detailed balance limit simulations, I reviewed material on the more complex aspects of IBSCs to prepare for the more detailed simulations using Simudo. After installing and testing Simudo I began by recreating previous results to ensure my setup was functioning correctly. A simulation conducted by a previous undergraduate student produced a table of values similar to Table II, using the same setup in Table I. The simulation results differ from the results found in [9] as the global mismatch vales are set to different percentages, and the optical cross-section values are reduced by a factor of ten.

The resulting values are presented in Table II for comparison, showing very little deviation from the previous values. These results were notably only performed at $\mu_I = 100$, simply to ensure accuracy. The accuracy of these results can be improved by decreasing the optimization tolerance in Simudo, at the expense of the time required to compute the values. The results in Table II took approximately 17 hours to complete with a tolerance of $\pm 1\%$. Future iterations could certainly improve the accuracy now that the simulation has demonstrated convergence.

TABLE I. Simulation Parameters

Value	Definition
$X = 1000$	Solar concentration
$\sigma_{CI}^{opt} = \sigma_{IV}^{opt}$	Optical cross-section
$\sigma_{CI}^{th} = \sigma_{IV}^{th} = 1.2 \times 10^{-15} cm^2$	Thermal cross-section
$v_{CB}^{th} = v_{IV}^{th} = 2 \times 10^7$	Thermal velocity
$N_I = 1 \times 10^{17}$	IB density of states
$E_I = [0.58, 0.55, 0.49, 0.37]$	IB energies (0-50%)
$E_g = 1.67\ eV$	Conduction band gap
$V_c = N_v = 5 \times 10^{18} cm^{-3}$	CB and VB density
$\mu_n = \mu_p = cm^2 V^{-1} s^{-1}$	Carrier mobility
$\mu_I = [0.001, 1, 30, 100]$	IB Carrier mobility
$\alpha_{cv} = 1 \times 10^4 cm^{-1}$	Absorption coeff.
$\epsilon_r = 13$	Relative permittivity
$T = 300$	Temp. of solar cell
$T_{sun} = 6000$	Temp. of sun
$N_A = N_D = 10^{17}$	Doping conc.
$NFSF = 1 \times 10^{19}$	Front surface field
$f_0 = 0.5$	Initial filling fraction

TABLE II. Comparison of Previous and New Results

Condition	Prev	New	Diff
0% GCM, $\sigma_{IV} = 5\sigma_{CI}$	30.6%	30.2%	0.7%
0% GCM, $\sigma_{IV} = \sigma_{CI}$	28.9%	28.9%	0.0%
0% GCM, $5\sigma_{IV} = 5\sigma_{CI}$	39.0%	38.7%	0.5%
0% GCM, $5\sigma_{IV} = \sigma_{CI}$	32.4%	32.0%	0.6%
10% GCM, $\sigma_{IV} = 5\sigma_{CI}$	30.2%	29.8%	0.7%
10% GCM, $\sigma_{IV} = \sigma_{CI}$	29.1%	28.7%	0.7%
10% GCM, $5\sigma_{IV} = 5\sigma_{CI}$	38.6%	38.3%	0.5%
10% GCM, $5\sigma_{IV} = \sigma_{CI}$	33.2%	32.8%	0.6%
25% GCM, $\sigma_{IV} = 5\sigma_{CI}$	29.5%	-	-
25% GCM, $\sigma_{IV} = \sigma_{CI}$	28.5%	28.1%	0.7%
25% GCM, $5\sigma_{IV} = 5\sigma_{CI}$	37.3%	36.9%	0.5%
25% GCM, $5\sigma_{IV} = \sigma_{CI}$	33.8%	33.4%	0.6%
50% GCM, $\sigma_{IV} = 5\sigma_{CI}$	27.9%	27.6%	0.6%
50% GCM, $\sigma_{IV} = \sigma_{CI}$	27.1%	26.8%	0.6%
50% GCM, $5\sigma_{IV} = 5\sigma_{CI}$	33.9%	33.6%	0.4%
50% GCM, $5\sigma_{IV} = \sigma_{CI}$	32.4%	32.1%	0.5%

Building off of the results of the previous student, the effects of IB mobility must be demonstrated at different CB energies, hence different E_g values, in order to prove that the effect is globally effective. To achieve this, the same parameters as seen in Table I are used, with the exception of the band gap E_g being set to 2.5 eV and the IB energies being modified to the values in Table III, to maintain the same m_g values.

TABLE III. IB energies for different mismatch values

m_g	E_I
0%	0.9739
10%	0.9327
25%	0.8622
50%	0.7107

The results of this simulation are presented in Table V, with notable missing data. The missing column at $\mu_I = 30$ is a result of my own mistake in overwriting the data after the simulation completed, while the incomplete horizontal rows are a result of convergence issues within Simudo. The optimization process requires computation of extremely large values on the order of 10^{25} and extremely small values on the order of 10^{-32} and smaller. Without increased stability, the computation can easily face difficulties with overflow in calculations.

As another comparison to previous results, four simulations were run by the previous student, leading to the values in Table IV. These values fit quite well with the simulated results, again with error attributed to personal mistakes and inaccurate optimization tolerance. $E_g = 2.5\ eV, GCM = 0\%\ \sigma_{ci} = \sigma_{iv} = 10^{-13} cm^2$

TABLE IV. Previous vs New Efficiency

Condition	Previous Eff.	New Eff.	Diff
$\mu_I = 0.001$	31.6%	31.1%	0.8%
$\mu_I = 1$	31.4%	30.4%	1.6%
$\mu_I = 30$	28.9%	-	-
$\mu_I = 100$	28.8%	28.1%	0.5%

TABLE V. Simulation at $E_g = 2.5$

Condition	0.001	1	30	100
$m_g = 0\%\ \sigma_{IV} = 5\sigma_{CI}$	29.3	32.8	-	29.0
$m_g = 0\%,\ \sigma_{IV} = \sigma_{CI}$	31.1	30.4	-	28.1
$m_g = 10\%\ \sigma_{IV} = 5\sigma_{CI}$	29.1	32.0	-	28.2
$m_g = 10\%,\ \sigma_{IV} = \sigma_{CI}$	29.5	29.3	-	27.4
$m_g = 10\%\ 5\sigma_{IV} = 5\sigma_{CI}$	42.3	42.5	-	41.6
$m_g = 25\%\ \sigma_{IV} = 5\sigma_{CI}$	28.4	30.3	-	26.7
$m_g = 25\%,\ \sigma_{IV} = \sigma_{CI}$	27.4	27.4	-	26.0
$m_g = 25\%,\ 5\sigma_{IV} = 5\sigma_{CI}$	39.8	39.8	-	39.0
$m_g = 50\%\ \sigma_{IV} = 5\sigma_{CI}$	25.8	26.1	-	23.7
$m_g = 50\%,\ \sigma_{IV} = \sigma_{CI}$	23.8	23.9	-	23.1
$m_g = 50\%\ 5\sigma_{IV} = 5\sigma_{CI}$	34.0	34.1	-	33.4
$m_g = 50\%,\ 5\sigma_{IV} = \sigma_{CI}$	30.9	32.1	-	32.3

The results of this simulation demonstrate the expected trend seen in [9]. When the optical cross-section of the CI process is much larger than that of the IV process, the efficiency of the solar cell is significantly impeded. The complete values were used to create the plot in Figure 9. This plot clearly demonstrates the trends in efficiency as μ_I values as the mobility increases, showing an unexpected decreasing trend in efficiency.

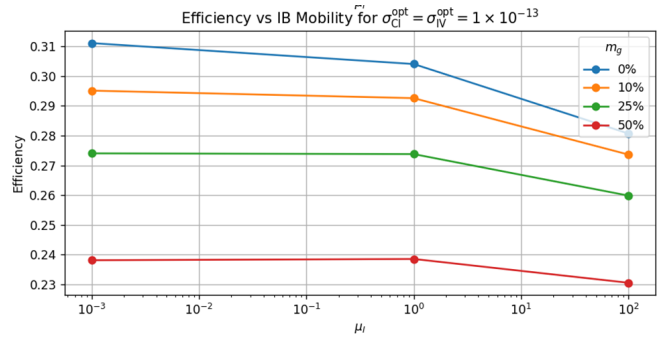

FIG. 9. Efficiency vs IB Mobility for $\sigma_{CI}^{opt} = \sigma_{IV}^{opt} = 1 \times 10^{-13}$.

The plot in Figure 10 shows a decrease in efficiency as the mobility deviates from $\mu_I = 1$ although the decrease in efficiency is expected with lower μ_I values. The decreased efficiency at higher μ_I values is unexpected.

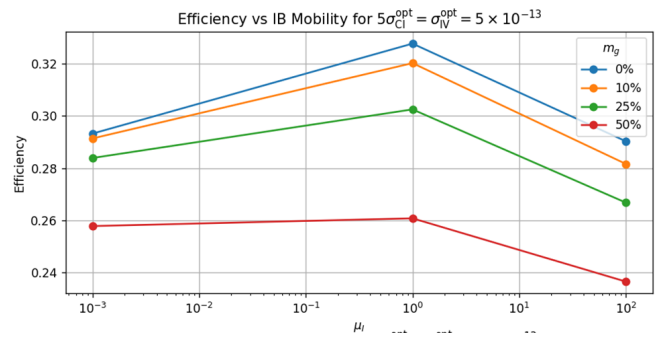

FIG. 10. Efficiency vs IB Mobility for $5\sigma_{CI}^{opt} = \sigma_{IV}^{opt} = 5 \times 10^{-13}$.

Overall, these results will require further analysis once the data is complete, although the trends in these plots are comparable to previous results in [9]. The experimental values may require review to ensure that inaccuracies are simply the result of tolerances within the simulation and not a separate factor.

VI. FUTURE ENDEAVOURS

The need for completion of Table V presents a clear objective to begin next semester. The collected data can then be analyzed to characterize any differences with previous results and present a trajectory for future exploration. I am interested in increasing the scope of the variables covered such as further decreasing the μ_I value to establish a clear trend and creating data point between the current values to potentially create models of the dependence of the efficiency on the IB mobility.

There are a plethora of parameters that can be altered to determine their influence on the IB mobility impact,

however I am limited in time. By defining two definite E_g values demonstrating a dependence on μ_I, the theory can be assumed to be demonstrable at most E_g values. This leaves opportunity to demonstrate if IB mobility still has major implications when various parameters are altered. It has already been shown that the IB mobility has varying impact when various optical cross-sections are used. The relationship between density of states in the IB, thermal cross-section and many other parameters can be studied.

VII. CONCLUSION

This semester has presented an excellent opportunity to become involved with practical research in the field of solar cells. This research could greatly benefit future designs of IBSCs and ensure that solar cells do not fall victim to unforeseen consequences of limited mobility within the IB. The results found thus far are minimal and require completion and significant elaboration. The trends do point to similarities with previous results and allow for comparable evaluations of the influence of the IB mobility. Next semester I will focus on continuously producing and analysing data in order to establish demonstrable trends in efficiency. This will require continuous simulation and hopefully an increase in the number of data points tested.

ACKNOWLEDGMENTS

I would like to sincerely thank Dr. Jacob Krich for sharing his expertise on the subject with a mastery and grace I will never fully comprehend. His explanation of complex topics such as this one is beyond wonder. I would also like to thank the other creators of Simudo, Eduard Dumitrescu and Matthew Wilkins, without their brilliantly designed framework and well commented code I would still be staring at an impressive and hopeless selection of errors.

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